

Project Planning & Management

Project Proposal:

Project Title

Land Type Classification Using Sentinel-2 Satellite Images with Deep Learning

1. Introduction

Land type classification plays an essential role in urban planning, environmental protection, precision agriculture, and natural resource management. Traditionally, identifying land cover from satellite imagery requires extensive manual analysis by experts, making the process time-consuming and costly.

With the rapid development of Artificial Intelligence and Deep Learning, satellite images can now be analysed automatically with high accuracy. This project aims to develop an intelligent system capable of classifying land types from Sentinel-2 multispectral satellite imagery using Deep Neural Networks (DNNs). The system will identify various land categories such as agriculture, urban areas, water bodies, forests, roads, and desert regions.

2. Problem Statement

Accurate land classification is essential for many governmental and environmental applications. However, manual interpretation of satellite imagery is labour-intensive, expensive, and prone to human error. Existing automated approaches may struggle with varying image quality, seasonal changes, and complex terrain.

This project addresses these challenges by developing a deep learning model capable of automatically classifying land types from Sentinel-2 satellite images with high accuracy and efficiency.

3. Project Objectives

The primary objectives of this project are:

- Collect and preprocess Sentinel-2 satellite imagery.
- Explore and analyse multispectral image data.
- Develop a deep learning model for land type classification.
- Evaluate model performance using standard classification metrics.
- Deploy the trained model as a web-based application or REST API.
- Provide an intuitive interface for users to upload satellite images and receive land type predictions.

4. Project Scope

In Scope

- Download and preprocess Sentinel-2 or EuroSAT datasets.
- Perform exploratory data analysis.
- Train and evaluate CNN-based deep learning models.
- Apply data augmentation and feature engineering techniques.
- Compare different deep learning architectures.
- Deploy the final model.
- Develop a simple interface for image upload and prediction.

Out of Scope

- Real-time satellite image acquisition.
- Satellite image generation.
- GIS editing and mapping beyond basic visualization.
- Commercial deployment.

5. Expected Outcomes

At the completion of the project, the system is expected to:

- Accurately classify satellite images into predefined land categories.
- Achieve competitive classification accuracy using deep learning.
- Provide a user-friendly interface for land classification.
- Assist researchers, planners, and environmental organizations in analysing land cover efficiently.

Project Plan

Project Overview

The project follows a collaborative team structure consisting of **six members**, organized into **three teams of two members each**. Each team is responsible for a specific phase of the project while maintaining continuous communication with the other teams to ensure smooth integration of all project components.

Regular meetings are scheduled throughout the project to discuss progress, share completed work, identify challenges, and provide support when needed. Although responsibilities are divided, all team members contribute to solving problems and making project decisions collaboratively.

Team Structure and Responsibilities

Team 1 – Data Collection, Preprocessing, and Exploration

Members:

- Nadine Tarek
- Nada Tarek

Responsibilities

- Collect Sentinel-2 satellite images.
- Explore the dataset and perform Exploratory Data Analysis (EDA).
- Analyse relationships between spectral bands.
- Clean the dataset.
- Report that explains data.

Team 2 – Data Analysis and Model Selection

Members:

- Ahmed Tarek
- Mahmoud Mohamed

Responsibilities

- Perform dimensionality reduction when appropriate.
- Apply preprocessing.
- Apply normalization.
- Apply feature engineering techniques such as normalization and data augmentation.
- Split the dataset into training, validation, and testing sets.

Team 3 – Model Development and Training

Members:

- Ashraquat Mostafa
- Youssef Mohamed

Responsibilities

- Develop the deep learning model.
- Train and validate the selected model.
- Select the most suitable model for land classification.
- Evaluate model performance.
- Generate performance metrics and visualizations.

Shared Responsibilities

The following phases are completed collaboratively by all team members:

- Model deployment
- System integration
- Testing and validation
- Monitoring and maintenance
- Documentation
- Final report preparation
- Presentation preparation
- Project updates and future improvements

Every member contributes to these activities to ensure consistency and quality across the entire system.

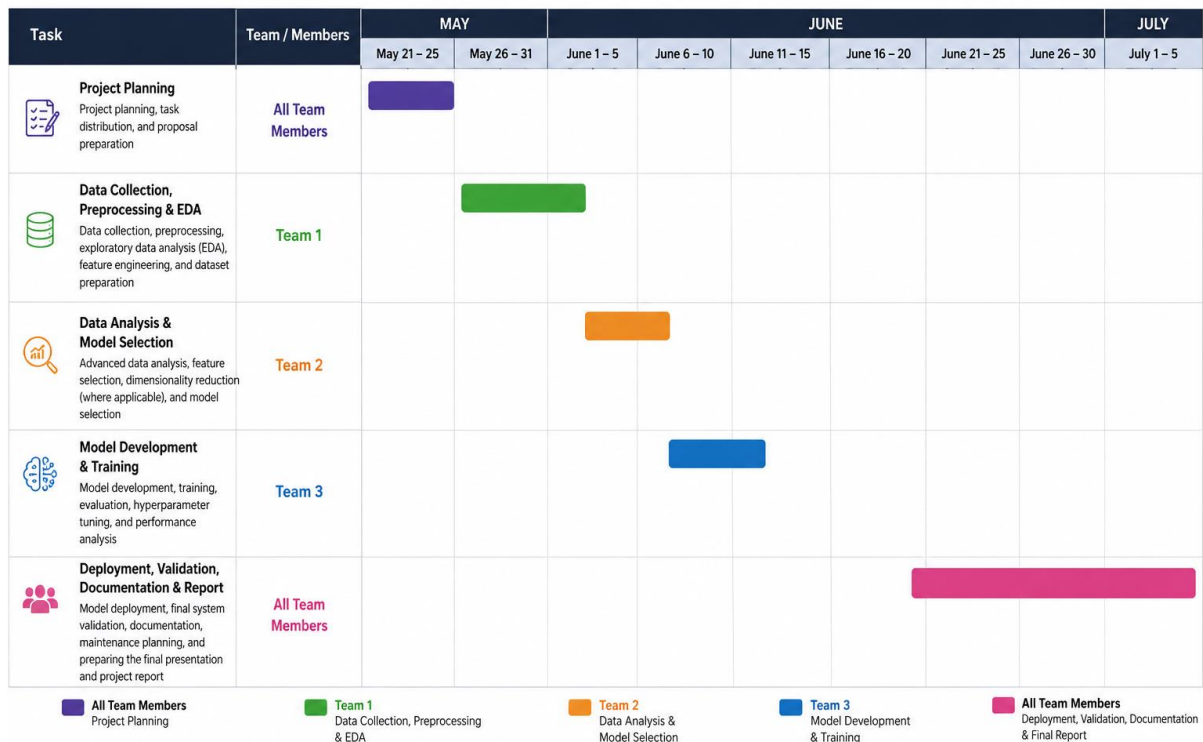
Project Timeline

The project was scheduled over a six-week period leading up to the final submission in July. The work was divided among three teams, with each team focusing on a specific milestone while maintaining continuous communication through regular meetings.

- **May 21 – May 25:** Project planning, task distribution, and proposal preparation by all team members.
- **May 26 – May 31: Team 1** carried out data collection, preprocessing, exploratory data analysis (EDA), feature engineering, and dataset preparation.
- **June 1 – June 5: Team 2** performed advanced data analysis, feature selection, dimensionality reduction where applicable, and evaluated different deep learning architectures to select the most suitable model.
- **June 6 – June 10: Team 3** developed, trained, and evaluated the deep learning model, including hyperparameter tuning and performance analysis.
- **June 25 – July 5:** The entire team focused on model deployment, final system validation, documentation, maintenance planning, and preparing the final presentation and project report.

Throughout the project, regular meetings were held where each team presented its progress, discussed challenges, and received feedback from the rest of the team. This collaborative approach ensured that technical issues were resolved efficiently and that all project components were successfully integrated.

PROJECT GANTT CHART



Resource Allocation

Each team focuses on its assigned phase while collaborating closely with the other teams. This division of responsibilities allows parallel development, improves productivity, and ensures that knowledge is shared among all members throughout the project.

Milestones, Responsibilities, and Deliverables

Milestone 1: Project Planning and Proposal

Duration: May 21 – May 25

Responsible Members:

- All Team Members

Activities:

- Define the project objectives and scope.
- Assign team roles and responsibilities.
- Prepare the project proposal and project plan.
- Develop the project timeline and communication strategy.

Deliverables:

- Project Proposal
- Project Plan
- Team Roles and Responsibilities

- Risk Assessment and Mitigation Plan
- Key Performance Indicators (KPIs)

Milestone 2: Data Collection, Preprocessing, and Exploration

Duration: May 26 – May 31

Responsible Members:

- **Nadine Tarek**
- **Nada Tarek**

Activities:

- Collect Sentinel-2 satellite images.
- Explore the dataset using Exploratory Data Analysis (EDA).
- Clean and preprocess the dataset.
- Apply feature engineering techniques such as normalization and data augmentation.
- Split the dataset into training, validation, and testing sets.

Deliverables:

- Cleaned Dataset
- EDA Report
- Feature Engineering Report
- Data Visualizations
- Prepared Training, Validation, and Test Datasets

Milestone 3: Data Analysis and Model Selection

Duration: June 1 – June 5

Responsible Members:

- **Ahmed Tarek**
- **Mahmoud Mohamed**

Activities:

- Analyze relationships between spectral bands.
- Perform dimensionality reduction where applicable.
- Compare different deep learning architectures.
- Select the most suitable model for land classification.
- Prepare model selection reports and performance comparisons.

Deliverables:

- Data Analysis Report
- Feature Analysis Report

- Model Selection Summary
- Model Performance Comparison
- Visualization of Analysis Results

Milestone 4: Model Development and Training

Duration: June 6 – June 10

Responsible Members:

- **Ashraquat Mostafa**
- **Youssef Mohamed**

Activities:

- Develop the deep learning model.
- Train and validate the selected model.
- Perform hyperparameter tuning.
- Evaluate model performance.
- Generate performance metrics and visualizations.

Deliverables:

- Source Code
- Trained Deep Learning Model
- Hyperparameter Tuning Results
- Model Evaluation Report
- Performance Metrics and Visualizations

Milestone 5: Deployment, Integration, and Finalization

Duration: June 25 – July 5

Responsible Members:

- **Nadine Tarek**
- **Nada Tarek**
- **Ahmed Tarek**
- **Mahmoud Mohamed**
- **Ashraquat Mostafa**
- **Youssef Mohamed**

Activities:

- Deploy the trained model.
- Develop the API and integrate system components.
- Perform testing and validation.
- Monitor and maintain the deployed system.
- Prepare the project documentation and final report.

- Prepare and deliver the final presentation.
- Plan future system updates and improvements.

Deliverables:

- Deployed Web Application/API
- System Integration
- Testing and Validation Report
- Technical Documentation
- User Manual
- Final Project Report
- Project Presentation
- Maintenance and Future Improvement Plan

Risk Assessment & Mitigation Plan

To ensure the successful completion of the project, the team identified several potential risks and established appropriate mitigation strategies to reduce their impact.

- **Poor quality or insufficient satellite images:** Low-quality or incomplete datasets may reduce the accuracy of the classification model. To address this, reliable datasets such as Sentinel-2 and EuroSAT will be used, and preprocessing techniques including data cleaning and augmentation will be applied.
- **Imbalanced dataset:** An uneven distribution of land classes may cause the model to perform poorly on underrepresented categories. This risk will be mitigated by using data augmentation techniques and maintaining balanced training, validation, and testing datasets.
- **Model overfitting:** The model may perform well on the training data but poorly on unseen data. Regularization techniques, dropout layers, early stopping, and cross-validation will be implemented to improve generalization.
- **Computational resource limitations:** Training deep learning models requires significant computational power, which may increase training time. The team will utilize Google Colab GPU or other available hardware resources and optimize the model to reduce computational costs.
- **Integration challenges:** Since different teams are responsible for different project phases, integrating all components may present technical challenges. Regular meetings, continuous communication, and frequent integration testing will help ensure smooth system integration.
- **Project schedule delays:** Delays in completing assigned tasks could affect the overall timeline. Progress will be monitored through scheduled meetings, and tasks will be redistributed among team members if necessary.

- **Deployment issues:** Unexpected errors may occur during system deployment. Comprehensive testing and validation will be conducted before deployment to identify and resolve potential issues.
- **Version control conflicts:** Simultaneous contributions from multiple team members may lead to code conflicts. GitHub will be used with a structured branching strategy, frequent commits, and regular code reviews to maintain consistency.

Key Performance Indicators (KPIs)

The success of the project will be evaluated using the following Key Performance Indicators:

- Achieve a **model accuracy of at least 80%** on the testing dataset.
- Obtain **precision, recall, and F1-score values of at least 80%**, ensuring reliable classification across all land categories.
- Maintain an **API response time of less than three seconds** for image classification requests.
- Ensure **system availability of at least 99%** during testing and demonstration.
- Successfully classify **at least 80% of uploaded satellite images** with accurate predictions.
- Complete **100% of the required project documentation** before the final submission deadline.
- Maintain an **up-to-date GitHub repository** with regular commits and documented progress throughout the project.
- Ensure **active participation of all team members** through regular meetings, collaboration, and timely completion of assigned tasks.